UNIT 1 · ARS/NET PATTERN

MCQ Bank — 60 Questions with Answers

Questions 1–60

Q1. The average farm power availability in India in 2021–22 was approximately:

(a) 1.84 kW/ha (b) 2.02 kW/ha
(c) 3.04 kW/ha (d) 4.0 kW/ha

Q2. The Government of India target for farm power availability by 2030 is:

(a) 3.5 kW/ha (b) 4.0 kW/ha
(c) 4.5 kW/ha (d) 5.0 kW/ha

Q3. The single largest source of farm power in India at present is:

(a) Electric motors (b) Diesel engines
(c) Tractors (d) Draught animals

Q4. Overall level of farm mechanization in India is about:

(a) 25–30% (b) 45–47%
(c) 60–65% (d) 75–80%

Q5. SMAM was launched in:

(a) 2010–11 (b) 2014–15
(c) 2016–17 (d) 2018–19

Q6. The Central Farm Machinery Training and Testing Institute is located at:

(a) Hisar (b) Budni
(c) Anantapur (d) Bhopal

Q7. Power threshers and chaff cutters are regulated for operator safety under:

(a) Factories Act 1948 (b) Dangerous Machines (Regulation) Act 1983
(c) Insecticides Act 1968 (d) BIS Act 1986

Q8. Which operation has the LOWEST mechanization level in India?

(a) Seedbed preparation (b) Wheat harvesting
(c) Rice transplanting & cotton picking (d) Irrigation

Q9. Complete inversion of the furrow slice is the distinguishing action of:

(a) Disc plough (b) Chisel plough
(c) Mouldboard plough (d) Cultivator

Q10. The part of the MB plough that absorbs side thrust is the:

(a) Frog (b) Share
(c) Landside (d) Shin

Q11. Bar-point share is preferred in:

(a) Sticky soils (b) Abrasive/stony soils causing rapid point wear
(c) Sandy loams (d) Peat soils

Q12. A stubble-type mouldboard is characterized by:

(a) Long, gently curved surface (b) Short, abruptly curved surface for maximum pulverization
(c) Slats (d) Very slender shape for high speed

Q13. Vertical suction in an MB plough helps in:

(a) Maintaining furrow width
(b) Penetration to proper depth
(c) Inversion
(d) Reducing draft

Q14. In a standard disc plough, the disc angle usually ranges between:

(a) 15–25°
(b) 30–35°
(c) 42–45°
(d) 50–60°

Q15. Penetration in a disc plough is obtained mainly by:

(a) Suction
(b) Weight of the plough
(c) Hydraulic down-pressure only
(d) Sharpness of share

Q16. Which plough has no landside?

(a) MB plough
(b) Standard disc plough
(c) Turn-wrest plough
(d) Victory plough

Q17. Continuous ploughing without back furrows and dead furrows is achieved with:

(a) Gathering method
(b) Casting method
(c) Two-way (reversible) plough
(d) Tandem harrow

Q18. A subsoiler normally operates at a depth of:

(a) 10–15 cm
(b) 20–30 cm
(c) 45–90 cm
(d) 100–150 cm

Q19. The blade type most common on Indian rotavators for trashy conditions is:

(a) C-type
(b) J-type
(c) L-type
(d) Pick type

Q20. An offset disc harrow is most suited for:

(a) Puddling
(b) Orchards — working under tree canopies
(c) Deep tillage
(d) Ridging

Q21. Laser land levelling typically saves irrigation water to the extent of:

(a) 5–10%
(b) 20–30%
(c) 50–60%
(d) 70–80%

Q22. Draft of an implement equals:

(a) Pull × sin θ
(b) Pull × cos θ
(c) Pull × tan θ
(d) Pull ÷ cos θ

Q23. Which seed-metering mechanism is most common on Indian seed drills?

(a) Cup feed
(b) Internal double run
(c) Fluted roller
(d) Vacuum disc

Q24. Compared with the internal double-run feed, the fluted-wheel feed is:

(a) Less positive in metering
(b) More positive/uniform
(c) Identical
(d) Unsuitable for wheat

Q25. Which furrow opener is best for trash-covered, minimum-till fields?

(a) Shoe
(b) Full runner
(c) Single-disc
(d) Shovel

Q26. In a seed-cum-fertilizer drill, fertilizer is placed relative to seed at about:

(a) Same level, mixed
(b) 5 cm below and 5 cm to the side
(c) 10 cm above
(d) 15 cm below only

Q27. Happy Seeder is used for:

(a) Transplanting rice
(b) Direct drilling of wheat into combine-harvested paddy residue
(c) Planting sugarcane setts
(d) Sowing on ridges

Q28. Mat-type nursery is a prerequisite for:

(a) Potato planter
(b) Self-propelled rice transplanter
(c) Sugarcane cutter planter
(d) Pneumatic planter

Q29. Standard row-to-row spacing of Japanese-type mat rice transplanters is:

(a) 150 mm
(b) 200 mm
(c) 238 mm
(d) 300 mm

Q30. Sugarcane cutter-planter plants:

(a) Whole cane
(b) 2–3 bud setts cut in the furrow
(c) Seedlings
(d) Single-bud chips only

Q31. The picker-wheel type potato planter's main drawback is:

(a) Low capacity
(b) Tuber damage by piercing picks
(c) Cannot form ridges
(d) High draft

Q32. Inclined-plate metering is preferred for groundnut because:

(a) Higher seed rate
(b) Minimum seed (kernel) damage
(c) Cheaper
(d) No calibration needed

Q33. Fertilizer hoppers require agitators and corrosion-resistant materials because fertilizers are:

(a) Volatile
(b) Hygroscopic and corrosive
(c) Explosive
(d) Oily

Q34. Spinning-disc broadcaster distributes fertilizer in a pattern best described as:

(a) Uniform rectangular
(b) Triangular, needing overlap
(c) Ring shaped
(d) Random

Q35. Cono-weeder is used in:

(a) Dryland wheat
(b) Puddled paddy between rows
(c) Sugarcane earthing
(d) Cotton topping

Q36. Sweep-type cultivators are favoured in dryland inter-culture because they:

(a) Invert soil
(b) Cut weeds below surface with least moisture loss
(c) Work deep
(d) Form ridges

Q37. For herbicide application on flat surfaces, the recommended nozzle is:

(a) Hollow cone
(b) Flat fan
(c) Solid stream
(d) Rotary

Q38. Doubling the spraying pressure increases nozzle discharge approximately:

(a) 2 times
(b) 4 times
(c) 1.41 times
(d) No change

Q39. ULV spraying refers to application rates of:

(a) <5 L/ha
(b) <50 L/ha
(c) <150 L/ha
(d) <400 L/ha

Q40. The ratio used to express droplet spectrum uniformity is:

(a) NMD/VMD
(b) VMD/NMD
(c) VMD × NMD
(d) VMD - NMD

Q41. In mechanical agitation of sprayer tanks, paddles usually rotate at:

(a) 10–50 rpm
(b) 100–200 rpm
(c) 500–800 rpm
(d) 1000–1500 rpm

Q42. In hydraulic (jet) agitation, nozzles near the tank bottom are aimed:

(a) Vertically up
(b) About 30° above horizontal
(c) Horizontally
(d) 45° downward

Q43. Most efficient hydraulic agitation is obtained at:

(a) High pressure
(b) Low pressure
(c) Medium pressure only
(d) Independent of pressure

Q44. Which pump handles wettable-powder suspensions with least wear problems?

(a) Gear pump
(b) Diaphragm pump
(c) Roller pump
(d) Piston pump without liners

Q45. Droplet sizes most prone to spray drift are below about:

(a) 400 µm
(b) 250 µm
(c) 100–150 µm
(d) 600 µm

Q46. The pitch of knife sections and guards in a conventional mower cutter bar is:

(a) 50.8 mm
(b) 76.2 mm
(c) 101.6 mm
(d) 88.9 mm

Q47. "Registration" of a mower knife means:

(a) Knife lead adjustment
(b) Section centre coinciding with guard centre at stroke ends
(c) Knife tension
(d) Serial number stamping

Q48. Lead given to the mower cutter bar is about:

(a) 2% of bar length
(b) 10% of bar length
(c) 25 mm regardless of length
(d) None

Q49. Cutting in a cutter-bar mower occurs by:

(a) Impact
(b) Shear between section and ledger plate
(c) Tearing
(d) Bending

Q50. Threshing action in conventional threshers is mostly due to:

(a) Rubbing only
(b) Impact
(c) Squeezing
(d) Centrifugal force

Q51. The cylinder type predominant in modern grain combines is:

(a) Spike tooth
(b) Rasp bar
(c) Wire loop
(d) Hammer mill

Q52. Recommended cylinder peripheral speed for threshing wheat is about:

(a) 5–10 m/s
(b) 12–15 m/s
(c) 25–30 m/s
(d) 40–50 m/s

Q53. Cylinder–concave clearance in a thresher/combine is kept:

(a) Uniform
(b) More at entry, less at exit
(c) Less at entry, more at exit
(d) Zero at entry

Q54. In an axial-flow paddy thresher, the crop:

(a) Passes straight through
(b) Moves spirally along the cylinder axis
(c) Falls by gravity only
(d) Is threshed by rolls

Q55. Safe feeding-chute dimensions of power threshers are specified in:

(a) IS 7898
(b) IS 9020
(c) IS 4468
(d) IS 6024

Q56. Reel index (reel speed ÷ forward speed) of a combine is normally:

(a) 0.8–1.0
(b) 1.25–1.5
(c) 2.5–3.0
(d) 4.0–5.0

Q57. In a rectangular baler, twine is tied by the:

(a) Plunger
(b) Knotter (billhook mechanism)
(c) Pickup
(d) Star wheel

Q58. The cotton spindle picker harvests:

(a) Whole plants
(b) Only seed cotton from open bolls (selective)
(c) Open + unopened bolls
(d) Lint only

Q59. Chopper-type sugarcane harvesters deliver cane as:

(a) Whole stalks
(b) 25–30 cm billets
(c) Bagasse
(d) Shredded trash

Q60. In an elevator-type potato digger, soil separation is achieved by:

(a) Blower
(b) Agitated rod-link chain conveyor
(c) Washing
(d) Sieves and fan

Answers

Q1 — (c) 3.04 kW/ha
Q2 — (b) 4.0 kW/ha
Q3 — (c) Tractors
Q4 — (b) 45–47%
Q5 — (b) 2014–15
Q6 — (b) Budni
Q7 — (b) Dangerous Machines (Regulation) Act 1983
Q8 — (c) Rice transplanting & cotton picking
Q9 — (c) Mouldboard plough
Q10 — (c) Landside
Q11 — (b) Abrasive/stony soils causing rapid point wear
Q12 — (b) Short, abruptly curved surface for maximum pulverization
Q13 — (b) Penetration to proper depth — *MB plough penetrates by suction; disc plough penetrates by weight — a classic contrast question.*
Q14 — (c) 42–45°
Q15 — (b) Weight of the plough — *MB plough penetrates by suction; disc plough penetrates by weight — a classic contrast question.*
Q16 — (b) Standard disc plough
Q17 — (c) Two-way (reversible) plough
Q18 — (c) 45–90 cm
Q19 — (c) L-type
Q20 — (b) Orchards — working under tree canopies
Q21 — (b) 20–30%
Q22 — (b) Pull × cos θ — *Draft is the horizontal, direction-of-travel component of pull.*

Q23 — (c) Fluted roller
Q24 — (b) More positive/uniform — *Kepner §11.6 — the fluted-wheel feed is "more positive in its metering action than the double run."*
Q25 — (c) Single-disc
Q26 — (b) 5 cm below and 5 cm to the side
Q27 — (b) Direct drilling of wheat into combine-harvested paddy residue
Q28 — (b) Self-propelled rice transplanter
Q29 — (c) 238 mm
Q30 — (b) 2–3 bud setts cut in the furrow
Q31 — (b) Tuber damage by piercing picks
Q32 — (b) Minimum seed (kernel) damage
Q33 — (b) Hygroscopic and corrosive
Q34 — (b) Triangular, needing overlap
Q35 — (b) Puddled paddy between rows
Q36 — (b) Cut weeds below surface with least moisture loss
Q37 — (b) Flat fan
Q38 — (c) 1.41 times — *$q \propto \sqrt{P}$, so doubling P multiplies q by $\sqrt{2} \approx 1.41$.*
Q39 — (a) <5 L/ha
Q40 — (b) VMD/NMD
Q41 — (b) 100–200 rpm
Q42 — (b) About 30° above horizontal — *Kepner §21.16 — jets 1–2 in. above the tank bottom, aimed 30° above horizontal.*
Q43 — (b) Low pressure — *Kepner §21.16 — low pressures give the most efficient hydraulic agitation.*
Q44 — (b) Diaphragm pump
Q45 — (c) 100–150 µm
Q46 — (b) 76.2 mm — *Standard cutter-bar geometry — 76.2 mm pitch.*
Q47 — (b) Section centre coinciding with guard centre at stroke ends — *Registration and alignment (lead) are different adjustments.*
Q48 — (a) 2% of bar length — *Registration and alignment (lead) are different adjustments; lead ≈ 2% of bar length.*
Q49 — (b) Shear between section and ledger plate — *Cutter-bar mowers cut by shear.*
Q50 — (b) Impact
Q51 — (b) Rasp bar
Q52 — (c) 25–30 m/s — *Kepner §17: 5000–6000 fpm ≈ 25–30 m/s for wheat.*
Q53 — (b) More at entry, less at exit — *A converging clearance grips the crop, then threshes it progressively without excess breakage.*
Q54 — (b) Moves spirally along the cylinder axis
Q55 — (b) IS 9020
Q56 — (b) 1.25–1.5
Q57 — (b) Knotter (billhook mechanism)
Q58 — (b) Only seed cotton from open bolls (selective)
Q59 — (b) 25–30 cm billets
Q60 — (b) Agitated rod-link chain conveyor